# Math 110 — Week 3

## Linear maps, null space and range, matrices, Markov chains (Notes Ch. 5–6, LADR 3A–3C, HW3)

## §1 Facts used from Weeks 1–2

(W1) Subspace test: $0 \in U$ and $\lambda v + w \in U$. Direct sum: every vector of $U_1 \oplus \cdots \oplus U_m$ is $u_1 + \cdots + u_m$ in exactly one way (Week 1, Prop. 7).

(W2) $v_1, \ldots, v_n$ is a basis of $V$ exactly when every $v \in V$ is $\lambda_1 v_1 + \cdots + \lambda_n v_n$ in exactly one way (Week 2).

(W3) An independent list extends to a basis; a spanning list contains a basis (Week 2, Props. 9, 11).

(W4) In a space of dimension $d$: $d$ independent vectors form a basis; a subspace of dimension $d$ is the whole space; a subspace of a finite-dimensional space is finite-dimensional (Week 2, Props. 13, 14).

(W5) Pigeonhole: a map between finite sets of equal size is injective exactly when it is surjective (Week 1).

(W6) Nonzero polynomials of different degrees are independent; $1, x, \ldots, x^m$ is a basis of $\mathcal{P}_m(\mathbb{R})$ (Week 2).

## §2 This week

### §2.1 Linear maps (Ch. 5.1)

**The question.** Which functions between vector spaces respect the two operations? For such a function, how much information is needed to know it completely? For example, knowing the derivatives of $1, x, x^2, x^3$, can we find the derivative of every cubic without differentiating it?

**Definition** (Def. 11, 12 / LADR 3.1, 5.1)**.** $T\colon V \to W$ is **linear** if $T(\lambda v + w) = \lambda T(v) + T(w)$ for all $v, w \in V$, $\lambda \in \mathbb{F}$. Equivalently $T(v + w) = Tv + Tw$ and $T(\lambda v) = \lambda Tv$; equivalently $T(\sum \lambda_k v_k) = \sum \lambda_k T(v_k)$. The set of linear maps $V \to W$ is $\mathcal{L}(V, W)$. A linear map $V \to V$ is an **operator**; $\mathcal{L}(V) = \mathcal{L}(V, V)$.

*Why the three forms agree.* 1. **Assume $T(v + w) = Tv + Tw$ and $T(\lambda v) = \lambda Tv$.** Then $T(\lambda v + w) = T(\lambda v) + T(w) = \lambda T(v) + T(w)$.

2. **Assume $T(\lambda v + w) = \lambda Tv + Tw$ for all $v, w, \lambda$.** With $\lambda = 1$ this is $T(v + w) = Tv + Tw$. With $v = w = 0$ and $\lambda = 1$ it gives $T(0) = 2T(0)$, so $T(0) = 0$. Then $w = 0$ gives $T(\lambda v) = \lambda T(v)$.

3. **Combinations of $n$ vectors.** Induction on $n$. The case $n = 1$ is $T(\lambda v) = \lambda Tv$. For $n \geq 2$,

$$T\Big(\sum_{k=1}^{n} \lambda_k v_k\Big) = T\Big(\lambda_n v_n + \sum_{k<n} \lambda_k v_k\Big) = \lambda_n T(v_n) + T\Big(\sum_{k<n} \lambda_k v_k\Big),$$

and the last term is $\sum_{k<n} \lambda_k T(v_k)$ by the induction hypothesis. Conversely, the form for $n = 2$ with $\lambda_2 = 1$ is the one-line form.

□

- Always $T(0) = 0$ (step 2). So a map with $T(0) \neq 0$ is not linear.
- Def. 13: the **restriction** of $T$ to a subspace $U$ is $T|_U\colon U \to W$, $T|_U(u) = T(u)$.

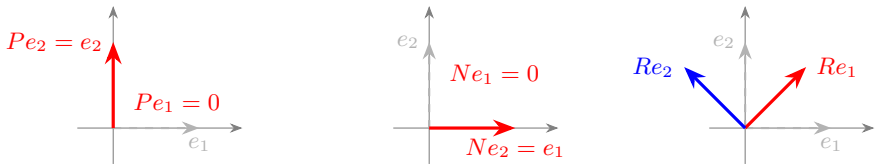
**Examples 21–23.** **(21)** Differentiation $p \mapsto p'$ on polynomials, and the map sending a periodic function to its sequence of Fourier coefficients, are linear. (Maps given by matrices are a third family; they appear with Def. 19.)

**(22)** The **shift** $S(a_0, a_1, a_2, \dots) = (a_1, a_2, a_3, \dots)$ on the space $\mathbb{R}^{\mathbb{N}_0}$ of real sequences $(a_0, a_1, \dots)$ (Week 1, Ex. 9). It maps the recurrence subspace $R = \{a : a_{n+2} = a_{n+1} + a_n\}$ into itself, and the restriction $S|_R \colon R \to R$ satisfies $(S|_R)^2 - S|_R - I = 0$, because $(S^2 a)_n = a_{n+2} = a_{n+1} + a_n = (Sa + a)_n$.

**(23)** Three maps on $\mathbb{R}^2$ used all semester: the projection $P(x, y) = (0, y)$; the map $N(x, y) = (y, 0)$, for which $Ne_2 = e_1$, $Ne_1 = 0$, so $N^2 = 0$; and rotation by $\theta$,

$$R_\theta(x, y) = (x\cos\theta - y\sin\theta,\ x\sin\theta + y\cos\theta).$$

Also $N = R_{-\pi/2}P$.




$P$ $\qquad$ $N$ $\qquad$ $R_{\pi/4}$

Where $P$, $N$ and $R_{\pi/4}$ send $e_1$ and $e_2$ (dashed: the original vectors). These images will be the columns of the three matrices in Section 2.3.

**Proposition** (Prop. 18, 19 / LADR 3.6–3.8)

$\mathcal{L}(V, W)$ is a vector space with $(\mu S + T)(v) = \mu S(v) + T(v)$. If $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$, the composition $ST$, $(ST)(u) = S(T(u))$, is linear.

*Proof.* 1. For $S, T$ linear and scalars $\lambda, \mu$, linearity of $S$ and of $T$ gives

$$(\mu S + T)(\lambda v + w) = \mu(\lambda Sv + Sw) + \lambda Tv + Tw = \lambda(\mu S + T)(v) + (\mu S + T)(w),$$

so $\mu S + T$ is linear. The zero map is linear. So $\mathcal{L}(V, W)$ is a subspace of the space of all functions $V \to W$.

2. Apply the linearity of $T$, then of $S$:

$$ST(\lambda u + v) = S(\lambda Tu + Tv) = \lambda STu + STv.$$

□

**Rules for composition.** Write $\mathrm{id}_V$ (or $I$) for the identity map $v \mapsto v$ of $V$. Each rule is checked by applying both sides to a vector.

1. Associativity: for $T_3 \in \mathcal{L}(U, V)$, $T_2 \in \mathcal{L}(V, W)$, $T_1 \in \mathcal{L}(W, X)$ and $u \in U$,
$$((T_1T_2)T_3)(u) = T_1(T_2(T_3u)) = (T_1(T_2T_3))(u).$$
2. Identities: for $T \in \mathcal{L}(V, W)$, $(T\,\mathrm{id}_V)(v) = T(v) = (\mathrm{id}_W\,T)(v)$.
3. $(S_1 + S_2)T(u) = S_1(Tu) + S_2(Tu)$, by the definition of the sum of two maps.
4. $S(T_1 + T_2)(u) = S(T_1u + T_2u) = ST_1u + ST_2u$. This rule uses that $S$ is linear.

Even when $ST$ and $TS$ are both defined, they can map between different spaces: for $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^3)$ and $S \in \mathcal{L}(\mathbb{R}^3, \mathbb{R}^2)$, $ST \in \mathcal{L}(\mathbb{R}^2)$ but $TS \in \mathcal{L}(\mathbb{R}^3)$. Composition is not commutative.

**Example 24: $\hat{P}\hat{X} - \hat{X}\hat{P} = I$.** On smooth functions let $(\hat{X}f)(x) = xf(x)$ and $\hat{P}f = f'$. Then $(\hat{P}\hat{X} - \hat{X}\hat{P})f = (xf)' - xf' = f$, so $\hat{P}\hat{X} - \hat{X}\hat{P} = I$.

**Theorem** (Prop. 21 / LADR 3.4: a linear map is determined by its values on a basis)

Let $v_1, \dots, v_n$ be a basis of $V$ and $w_1, \dots, w_n$ any vectors in $W$. There is exactly one linear map $T\colon V \to W$ with $T(v_j) = w_j$ for all $j$.

### The idea

Uniqueness comes first and tells us what the map must be. Every $v$ is $\lambda_1 v_1 + \cdots + \lambda_n v_n$, so linearity forces $T(v) = \lambda_1 w_1 + \cdots + \lambda_n w_n$. For existence, take that formula as the definition. The risk with "define $T(v)$ using a way of writing $v$" is that $v$ might be written in two ways that give different answers. Here it cannot: the coefficients $\lambda_k$ are unique because $v_1, \dots, v_n$ is a basis (W2). If the list were dependent the definition could contradict itself; if it did not span, $T$ would not be defined everywhere.

*Proof.* 1. *Uniqueness.* If $T$ is linear with $T(v_j) = w_j$ and $v = \sum \lambda_j v_j$, then $T(v) = \sum \lambda_j w_j$.

2. *Existence.* For $v = \sum \lambda_j v_j$ (unique coefficients, W2) define $T(v) = \sum \lambda_j w_j$. Then $T(v_j) = w_j$.

3. *Linear.* If $v = \sum \lambda_j v_j$ and $v' = \sum \lambda'_j v_j$, then $\mu v + v' = \sum (\mu\lambda_j + \lambda'_j) v_j$, so

$$T(\mu v + v') = \sum (\mu\lambda_j + \lambda'_j) w_j = \mu T(v) + T(v').$$

□

**Proposition** (Prop. 20: gluing maps on a direct sum)

If $U_1 \oplus \cdots \oplus U_m$ is a direct sum and $T_i\colon U_i \to W$ are linear, there is exactly one linear $T\colon U_1 \oplus \cdots \oplus U_m \to W$ with $T|_{U_i} = T_i$ for all $i$, given by

$$T(u_1 + \cdots + u_m) = T_1(u_1) + \cdots + T_m(u_m).$$

*Proof.* 1. *Uniqueness.* If $T$ is linear and equals $T_i$ on each $U_i$, then

$$T(u_1 + \cdots + u_m) = T(u_1) + \cdots + T(u_m) = T_1(u_1) + \cdots + T_m(u_m).$$

So the formula is the only possibility.

2. *The formula is a valid definition.* Every vector of the direct sum is $u_1 + \cdots + u_m$ with $u_i \in U_i$ in exactly one way (W1), so the formula assigns it exactly one value.

3. *Linear.* If $v = \sum u_i$ and $v' = \sum u'_i$, then $\lambda v + v' = \sum (\lambda u_i + u'_i)$ with $\lambda u_i + u'_i \in U_i$, so

$$T(\lambda v + v') = \sum T_i(\lambda u_i + u'_i) = \lambda \sum T_i(u_i) + \sum T_i(u'_i) = \lambda T(v) + T(v').$$

4. *Restrictions.* A vector $u \in U_i$ is the sum with $u$ in position $i$ and 0 elsewhere, so $T(u) = T_i(u)$.

Prop. 21 is the special case where $U_j$ is the line $\mathbb{F}v_j$. □

**The question answered.** The functions that respect the operations are the linear maps. A linear map on a space with basis $v_1, \dots, v_n$ is known completely once $T(v_1), \dots, T(v_n)$ are known, and every choice of these $n$ vectors comes from exactly one linear map (Prop. 21). For differentiation on $\mathcal{P}_3(\mathbb{R})$: $D(1) = 0$, $D(x) = 1$, $D(x^2) = 2x$, $D(x^3) = 3x^2$ give

$$D(a + bx + cx^2 + dx^3) = b + 2cx + 3dx^2.$$

### Shortcuts

- **Not linear? Check $T(0) = 0$ first.** Reason: every linear map has $T(0) = 0$. Condition: this test can only rule a map out; $T(0) = 0$ does not prove linearity. Instance: $T(x, y) = (x + 1, y)$ has $T(0, 0) = (1, 0)$, so it is not linear.
- **Proving linearity in one line.** Verify $T(\lambda v + w) = \lambda T(v) + T(w)$ (Def. 11). Instance: for $T(p) = p(1) + p'(0)$ on $\mathcal{P}_2(\mathbb{R})$, $(\lambda p + q)(1) + (\lambda p + q)'(0) = \lambda T(p) + T(q)$.

- **Defining a map, or proving two maps equal: use a basis.** Reason: Prop. 21. Condition: the list must be a basis. Instance: the linear $T$ on $\mathbb{R}^2$ with $T(1,1)=(1,0)$ and $T(1,-1)=(0,0)$ is $T(x,y)=(\frac{x+y}{2},0)$, because $(x,y)=\frac{x+y}{2}(1,1)+\frac{x-y}{2}(1,-1)$.

**Traps**

- $T(0)=0$ does not make $T$ linear: $T(x,y)=(x^2,y)$ has $T(0)=0$, but $T(2,0)=(4,0)\neq 2T(1,0)=(2,0)$.
- $ST\neq TS$ in general. With $P,N$ from Ex. 23, $PN(x,y)=P(y,0)=(0,0)$ but $NP(x,y)=N(0,y)=(y,0)$, so $PN=0\neq N=NP$.
- $ST=0$ does not give $S=0$ or $T=0$: $NN=0$ although $N\neq 0$.
- Prescribing values on a list that is not a basis can be impossible: no linear $T$ on $\mathbb{R}^2$ has $T(1,0)=(1,0)$ and $T(2,0)=(0,1)$, since linearity forces $T(2,0)=2T(1,0)=(2,0)$.

## §2.2 Null space and range (Ch. 5.2)

**The question.** A linear map can send different vectors to the same vector and can miss part of the target. How are the two measured, and how are they related? For $P(x,y)=(0,y)$: everything on the $x$-axis is sent to 0, and only the $y$-axis is reached.

**Definition** (Def. 14, 15 / LADR 3.11, 3.16)**.** For $T\in\mathcal{L}(V,W)$: the **null space** (kernel, written $\ker T$) is $\operatorname{null} T=\{v\in V : Tv=0\}$; the **range** (image, written $\operatorname{im} T$) is $\operatorname{range} T=\{Tv : v\in V\}$. $\dim\operatorname{null} T$ is the **nullity**, $\dim\operatorname{range} T$ the **rank**.

**Proposition** (Prop. 22, 23 / LADR 3.13, 3.15, 3.18, 3.19)

$\operatorname{null} T$ is a subspace of $V$ and $\operatorname{range} T$ is a subspace of $W$. $T$ is injective $\iff \operatorname{null} T=\{0\}$. $T$ is surjective $\iff \operatorname{range} T=W$.

**The idea**

For a general function, injectivity has to be checked for every pair of inputs. For a linear map, $Tv=Tw$ is the same as $T(v-w)=0$, so two inputs collide exactly when their difference is in the null space. Checking the single equation $Tv=0$ is enough.

*Proof.* 1. $T0=0$, so $0\in\operatorname{null} T$. If $Tv_1=Tv_2=0$ then $T(\lambda v_1+v_2)=0$.

2. $0=T0\in\operatorname{range} T$. If $w_1=Tv_1$, $w_2=Tv_2$ then $\lambda w_1+w_2=T(\lambda v_1+v_2)$.

3. If $T$ is injective and $Tv=0=T0$, then $v=0$. Conversely, if $\operatorname{null} T=\{0\}$ and $Tv=Tw$, then $T(v-w)=0$, so $v-w=0$.

4. Surjective means every $w$ is some $Tv$, which is $\operatorname{range} T=W$.

□

**Example 25: null space and range of differentiation.** Let $D$ be differentiation on $\mathcal{P}_4(\mathbb{R})$. $\operatorname{null} D=\operatorname{span}(1)$, since $p'=0$ exactly for the constants. $\operatorname{range} D$ is spanned by the images of the basis $1,x,x^2,x^3,x^4$, which are

$$D(1)=0,\quad D(x)=1,\quad D(x^2)=2x,\quad D(x^3)=3x^2,\quad D(x^4)=4x^3,$$

so $\operatorname{range} D=\operatorname{span}(1,2x,3x^2,4x^3)=\mathcal{P}_3(\mathbb{R})$. Check with rank–nullity (next): $5=1+4$.

**Theorem** (Prop. 24 / LADR 3.21: fundamental theorem of linear maps (rank–nullity))

Let $V$ be finite-dimensional and $T \in \mathcal{L}(V, W)$. There is a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ of $V$ such that $u_1, \ldots, u_m$ is a basis of $\operatorname{null} T$ and $Tv_1, \ldots, Tv_n$ is a basis of $\operatorname{range} T$. In particular

$$\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T.$$

**The idea**

The range is spanned by the images of any basis of $V$. Some of those images may be 0 or redundant. To get a basis of the range, choose the basis of $V$ so that the vectors with image 0 are listed separately: start with a basis $u_1, \ldots, u_m$ of $\operatorname{null} T$ and extend it by $v_1, \ldots, v_n$. Then $\operatorname{range} T$ is spanned by $Tv_1, \ldots, Tv_n$ alone. A relation $\sum \lambda_j T v_j = 0$ says $\sum \lambda_j v_j$ is in the null space, so it is a combination of the $u$'s; since the $u$'s and $v$'s together are independent, all $\lambda_j = 0$. So no nonzero combination of the $v_j$ is in $\operatorname{null} T$.

*Proof.* 1. $\operatorname{null} T$ is finite-dimensional (W4). Choose a basis $u_1, \ldots, u_m$ of it and extend to a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ of $V$ (W3).

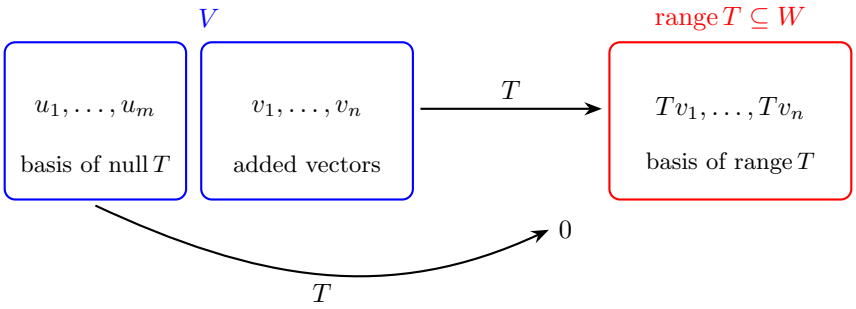
2. $Tv_1, \ldots, Tv_n$ *spans* $\operatorname{range} T$. Write any $v \in V$ as $v = \sum \mu_i u_i + \sum \lambda_j v_j$. Since every $Tu_i = 0$,
$$Tv = \sum \mu_i T u_i + \sum \lambda_j T v_j = \sum \lambda_j T v_j.$$

3. *Independent.* Suppose $\sum \lambda_j T v_j = 0$. Then $T(\sum \lambda_j v_j) = 0$, so $\sum \lambda_j v_j \in \operatorname{null} T$, and $\sum \lambda_j v_j = \sum \mu_i u_i$ for some $\mu_i$.

4. Then $\sum \mu_i u_i - \sum \lambda_j v_j = 0$ is a relation in a basis of $V$, so all $\lambda_j$ (and $\mu_i$) are 0.

5. Count: $\dim V = m + n = \dim \operatorname{null} T + \dim \operatorname{range} T$.

□




A basis of $V$ adapted to $T$. The first group is sent to 0. The images of the second group are a basis of the range. Counting: $\dim V = m + n$. For $D$ on $\mathcal{P}_4(\mathbb{R})$: null space basis 1, added vectors $x, \ldots, x^4$, images $1, 2x, 3x^2, 4x^3$.

**Proposition** (Prop. 25: equal dimensions)

If $\dim V = \dim W < \infty$ and $T \in \mathcal{L}(V, W)$, then $T$ injective $\iff$ $T$ surjective $\iff$ $T$ bijective. In particular this holds for every operator on a finite-dimensional space.

**The idea**

This is the pigeonhole principle (W5) with dimension in place of size. Rank–nullity gives the count: $\dim \operatorname{range} T = \dim V - \dim \operatorname{null} T$, so each extra dimension of $\operatorname{null} T$ lowers $\dim \operatorname{range} T$ by one.

*Proof.* 1. If $T$ is injective, $\dim \operatorname{null} T = 0$, so $\dim \operatorname{range} T = \dim V = \dim W$, so $\operatorname{range} T = W$ (W4).

2. If $T$ is surjective, $\dim \operatorname{range} T = \dim W = \dim V$, so $\dim \operatorname{null} T = 0$ and $T$ is injective.

□

- **Consequences of the count (LADR 3.22, 3.24).** $\dim \operatorname{range} T \le \min(\dim V, \dim W)$. If $\dim V > \dim W$, no linear map $V \to W$ is injective, since $\dim \operatorname{null} T = \dim V - \dim \operatorname{range} T \ge \dim V - \dim W > 0$. If $\dim V < \dim W$, none is surjective, since $\dim \operatorname{range} T \le \dim V < \dim W$.

- **Homogeneous systems.** The system $\sum_j a_{ij}x_j = 0$ $(i = 1, \dots, m)$ of $m$ equations in $n > m$ unknowns has a nonzero solution: the linear map $T(x_1, \dots, x_n) = \big(\sum_j a_{1j}x_j, \dots, \sum_j a_{mj}x_j\big)$ from $\mathbb{F}^n$ to $\mathbb{F}^m$ is not injective, and its null space is the solution set. A system of $m > n$ equations in $n$ unknowns has no solution for some right side: the same kind of map $\mathbb{F}^n \to \mathbb{F}^m$ is not surjective.
- **The formula is about dimensions only.** For $P(x, y) = (0, y)$: $\operatorname{null} P = \operatorname{span}(e_1)$, $\operatorname{range} P = \operatorname{span}(e_2)$, and $\mathbb{R}^2 = \operatorname{null} P \oplus \operatorname{range} P$. For $N(x, y) = (y, 0)$: $\operatorname{null} N = \operatorname{span}(e_1) = \operatorname{range} N$. Still $2 = 1 + 1$, but the null space and the range are the same line.

**The question answered.** Lost information is measured by $\operatorname{null} T$: two inputs have the same image exactly when their difference is in $\operatorname{null} T$, so $T$ is injective exactly when $\operatorname{null} T = \{0\}$. The part of the target that is reached is $\operatorname{range} T$. Rank–nullity ties the two together: $\dim \operatorname{null} T + \dim \operatorname{range} T = \dim V$. For $P(x, y) = (0, y)$: $\operatorname{null} P$ is the $x$-axis, $\operatorname{range} P$ is the $y$-axis, and $1 + 1 = 2$.

### Shortcuts

- **Rank–nullity does half the work.** Reason: Prop. 24. Condition: $V$ finite-dimensional. Instance: $T(x, y, z) = (x + y, y + z)$ from $\mathbb{R}^3$ to $\mathbb{R}^2$ is surjective ($T(1, 0, 0) = (1, 0)$, $T(0, 0, 1) = (0, 1)$), so $\dim \operatorname{null} T = 3 - 2 = 1$; indeed $\operatorname{null} T = \operatorname{span}\big((1, -1, 1)\big)$.
- **Dimension of a subspace cut out by $k$ conditions.** Write it as $\operatorname{null} S$ for a linear $S \colon V \to \mathbb{F}^k$; if $S$ is surjective, the dimension is $\dim V - k$. Reason: Prop. 24. Condition: show surjectivity by finding $k$ vectors whose images span $\mathbb{F}^k$. Instance: $\{p \in \mathcal{P}_3(\mathbb{R}) : p(0) = p(1) = 0\} = \operatorname{null} S$ for $S(p) = (p(0), p(1))$; $S(1 - x) = (1, 0)$ and $S(x) = (0, 1)$, so the dimension is $4 - 2 = 2$ (a basis is $x(x - 1)$, $x^2(x - 1)$).
- **Equal dimensions: check injectivity only.** Reason: Prop. 25. Condition: $\dim V = \dim W < \infty$. Instance: $T(p) = p + p'$ on $\mathcal{P}_2(\mathbb{R})$. If $p = a + bx + cx^2$ and $Tp = (a + b) + (b + 2c)x + cx^2 = 0$, then $c = 0$, $b = 0$, $a = 0$; so $T$ is bijective.
- **Test injectivity and surjectivity on a basis.** $T$ is injective exactly when $Tv_1, \dots, Tv_n$ is independent, and surjective exactly when it spans $W$. Reason: $\operatorname{range} T = \operatorname{span}(Tv_1, \dots, Tv_n)$, and $\sum \lambda_j Tv_j = 0$ exactly when $\sum \lambda_j v_j \in \operatorname{null} T$. Instance: $T(p) = (p(0), p(1), p(2))$ on $\mathcal{P}_2(\mathbb{R})$ sends $1, x, x^2$ to $(1, 1, 1), (0, 1, 2), (0, 1, 4)$, which are independent, so $T$ is injective.
- **Impossibility by counting.** Reason: $\dim \operatorname{range} T \le \dim W$ together with Prop. 24. Instances: no $T \colon \mathbb{R}^4 \to \mathbb{R}^2$ has $\dim \operatorname{null} T = 1$, since that would give rank $3 > 2$ (Midterm 1 review Ex. 6). For $T \in \mathcal{L}(V)$, $\operatorname{null} T = \operatorname{range} T$ forces $\dim V = 2 \dim \operatorname{null} T$, so $\dim V$ is even (HW3 P2).
- **Prescribed null space and range.** For subspaces $N \subseteq V$ and $R \subseteq W$ with $V$ finite-dimensional, some $T \in \mathcal{L}(V, W)$ has $\operatorname{null} T = N$ and $\operatorname{range} T = R$ exactly when $\dim N + \dim R = \dim V$. Reason: necessity is Prop. 24. For existence, extend a basis $u_1, \dots, u_m$ of $N$ to a basis $u_1, \dots, u_m, v_1, \dots, v_n$ of $V$, choose a basis $r_1, \dots, r_n$ of $R$, and set $Tu_i = 0$, $Tv_j = r_j$ (Prop. 21). Then $\operatorname{range} T = \operatorname{span}(r_j) = R$, and $\operatorname{null} T \supseteq N$ has dimension $\dim V - n = m$, so $\operatorname{null} T = N$ (W4). Instance: HW3 P2(a).

### Traps

- $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ does not say $V = \operatorname{null} T \oplus \operatorname{range} T$: for $N$, $\operatorname{null} N = \operatorname{range} N = \operatorname{span}(e_1)$.
- An independent list need not be sent to an independent list: $T = 0$ sends $e_1$ to 0. A dependent list is always sent to a dependent list (HW3 P4). Every independent list stays independent exactly when $T$ is injective: if $T$ is injective and $\sum \lambda_j Tv_j = 0$, then $T(\sum \lambda_j v_j) = 0$, so $\sum \lambda_j v_j = 0$; if $T$ is not injective, some $v \neq 0$ has $Tv = 0$, and the independent list $v$ is sent to the list 0.
- "Injective $\iff$ surjective" needs $\dim V = \dim W < \infty$. The shift $S$ of Ex. 22 is surjective, since $(b_0, b_1, \dots) = S(0, b_0, b_1, \dots)$, but not injective, since $S(1, 0, 0, \dots) = 0$. And $T(x) = (x, 0)$ from $\mathbb{R}$ to $\mathbb{R}^2$ is injective but not surjective.

## §2.3 Matrices (Ch. 6)

**The question.** For differentiation $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$ and the antiderivative $S\colon \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ with zero constant term, how do we record each map as a table of numbers, and how do we get the table of $SD$ from the two tables without computing $SD$ directly?

**Definition** (Def. 16, 17 / LADR 3.29, 3.31)**.** $\mathbb{F}^{m,n}$ is the set of $m \times n$ matrices $A = (a_{ij})$ ($m$ rows, $n$ columns) with entries in $\mathbb{F}$. From now on, elements of $\mathbb{F}^n$ are written as columns. Let $B = (v_1, \dots, v_n)$ be a basis of $V$ and $C = (w_1, \dots, w_m)$ a basis of $W$. The **matrix of** $T \in \mathcal{L}(V, W)$ is $\mathcal{M}(T) = (a_{ij}) \in \mathbb{F}^{m,n}$ (written $\mathcal{M}_B^C(T)$ when the bases must be named) where

$$T(v_j) = a_{1j}w_1 + \cdots + a_{mj}w_m.$$

**Column $j$ of $\mathcal{M}(T)$ is the coordinate vector of $T(v_j)$ in the basis $C$.** $\mathbb{F}^{m,n}$ is a vector space of dimension $mn$, with basis the matrices $E_{i,j}$ (a single 1 in row $i$, column $j$); Ex. 27: $E_{1,1}, E_{1,2}, E_{2,1}, E_{2,2}$ for $\mathbb{F}^{2,2}$.

**Example 26: matrix operations entry by entry.** In $\mathbb{C}^{2,3}$, addition and scalar multiplication are entry by entry:

$$\begin{bmatrix} 1+i & 3+2i & 4 \\ 5 & i & 2+i \end{bmatrix} + \begin{bmatrix} 4-2i & 5 & 1 \\ 3 & 2 & i \end{bmatrix} = \begin{bmatrix} 5-i & 8+2i & 5 \\ 8 & 2+i & 2+2i \end{bmatrix},$$

$$(1+i)\begin{bmatrix} 1+i & 3+2i & 4 \\ 5 & i & 2+i \end{bmatrix} = \begin{bmatrix} 2i & 1+5i & 4+4i \\ 5+5i & -1+i & 1+3i \end{bmatrix}.$$

In the standard basis, the images in the figure of Section 2.1 give $Pe_1 = 0$, $Pe_2 = e_2$; $R_\theta e_1 = (\cos\theta, \sin\theta)$, $R_\theta e_2 = (-\sin\theta, \cos\theta)$; $Ne_1 = 0$, $Ne_2 = e_1$. So

$$\mathcal{M}(P) = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathcal{M}(R_\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad \mathcal{M}(N) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

**Example 28: building a matrix column by column.** $T\colon \mathcal{P}_3(\mathbb{C}) \to \mathbb{C}^4$, $T(p) = (p(1), p(i), p(-1), p(-i))$, bases $1, x, x^2, x^3$ and $e_1, \dots, e_4$.

1. $T(1) = (1,1,1,1)$; $T(x) = (1, i, -1, -i)$; $T(x^2) = (1, -1, 1, -1)$; $T(x^3) = (1, -i, -1, i)$.
2. These are the columns:

$$\mathcal{M}(T) = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{bmatrix}.$$

**Proposition** (Prop. 26)

With bases fixed, $T \mapsto \mathcal{M}(T)$ is a bijection $\mathcal{L}(V, W) \to \mathbb{F}^{m,n}$, and $\mathcal{M}(S+T) = \mathcal{M}(S) + \mathcal{M}(T)$, $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$. Hence $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$.

*Proof.* 1. *Injective.* If $\mathcal{M}(S) = \mathcal{M}(T)$, then for each $j$ the vectors $Sv_j$ and $Tv_j$ have the same coordinates in $C$, so $Sv_j = Tv_j$. Two linear maps that agree on a basis are equal (Prop. 21).

2. *Surjective.* Given any matrix $A = (a_{ij})$, let $w_j' = a_{1j}w_1 + \cdots + a_{mj}w_m$. By Prop. 21 there is a linear $T$ with $Tv_j = w_j'$, and then $\mathcal{M}(T) = A$.

3. *Linear.* $(S+T)v_j = Sv_j + Tv_j$, so column $j$ of $\mathcal{M}(S+T)$ is the sum of the $j$-th columns. In the same way $(\lambda T)v_j = \lambda T v_j$ gives $\mathcal{M}(\lambda T) = \lambda \mathcal{M}(T)$.

4. *Dimension.* For each $i, j$ let $T_{ij} \in \mathcal{L}(V, W)$ be the map with $T_{ij}(v_j) = w_i$ and $T_{ij}(v_k) = 0$ for $k \neq j$ (Prop. 21), so $\mathcal{M}(T_{ij}) = E_{i,j}$. **They span.** If $\mathcal{M}(T) = (a_{ij})$, then $\mathcal{M}(T) = \sum a_{ij}E_{i,j} = \mathcal{M}\big(\sum a_{ij}T_{ij}\big)$ by step 3, so

$T = \sum a_{ij} T_{ij}$ by step 1. **They are independent.** If $\sum c_{ij} T_{ij} = 0$, then $\sum c_{ij} E_{i,j} = \mathcal{M}(0) = 0$, so every $c_{ij} = 0$. Hence $\dim \mathcal{L}(V, W) = mn$.

□

**Definition** (Def. 18, 19 / LADR 3.73, 3.76: coordinates and matrix times vector)**.** The **coordinates** of $v = \lambda_1 v_1 + \cdots + \lambda_n v_n$ in the basis $B$ form the column $\mathcal{M}(v)$ (written $\mathcal{M}_B(v)$ when the basis must be named) with entries $\lambda_1, \ldots, \lambda_n$; to save space we write it as $(\lambda_1, \ldots, \lambda_n)^{\mathsf{T}}$, where the $\mathsf{T}$ only means "written as a column". The product of a matrix and a column is defined so that

$$\mathcal{M}(T)\,\mathcal{M}(v) = \mathcal{M}(Tv).$$

In entries: if $A = (a_{ij})$ and $x = (\lambda_j)$, then $(Ax)_i = \sum_j a_{ij} \lambda_j$.

**The idea**

The product is what the coordinates of $Tv$ have to be. $Tv = \sum_j \lambda_j T(v_j)$, and $T(v_j)$ has coordinates "column $j$". So the coordinates of $Tv$ are $\lambda_1(\text{column } 1) + \cdots + \lambda_n(\text{column } n)$. Reading this row by row gives $(Ax)_i = \sum_j a_{ij}\lambda_j$. So $Ax$ is a combination of the columns of $A$ with the entries of $x$ as coefficients: the range of $x \mapsto Ax$ is the span of the columns, and $Ae_j$ is column $j$ of $A$.

For a fixed $A \in \mathbb{F}^{m,n}$, the map $x \mapsto Ax$ from $\mathbb{F}^n$ to $\mathbb{F}^m$ is linear, because $(Ax)_i = \sum_j a_{ij} x_j$ is linear in $x$.

**Null space, range and rank of a matrix.** A matrix $A \in \mathbb{F}^{m,n}$ gives the linear map $x \mapsto Ax$ from $\mathbb{F}^n$ to $\mathbb{F}^m$. The **null space**, **range** and **rank** of $A$ are those of this map. By the idea box, range $A$ is the span of the columns of $A$, so rank $A$ is the dimension of the span of the columns.

**Example 29: coordinates in two bases.** In $\mathcal{P}_2(\mathbb{R})$, $1 + 2x + 3x^2$ has coordinates $(1, 2, 3)$ in the basis $1, x, x^2$. In the basis $1 + x,\ 1 + x^2,\ x + x^2$, solve $a(1 + x) + b(1 + x^2) + c(x + x^2) = 1 + 2x + 3x^2$: the constant, $x$ and $x^2$ coefficients give $a + b = 1$, $a + c = 2$, $b + c = 3$, so $a = 0$, $b = 1$, $c = 2$, and the coordinates are $(0, 1, 2)$.

**Worked example: a matrix in bases that are not standard.** Let $T\colon \mathbb{R}^2 \to \mathbb{R}^2$, $T(a, b) = (a + b,\ a - b)$, with basis $B = ((1, 1), (1, -1))$ of the domain and $C = ((1, 0), (1, 1))$ of the target.

1. Apply $T$ to the vectors of $B$, using the formula for $T$ (do not choose the images): $T(1, 1) = (2, 0)$ and $T(1, -1) = (0, 2)$.
2. Write each image in $C$. $(2, 0) = 2(1, 0) + 0(1, 1)$. For $(0, 2) = s(1, 0) + t(1, 1)$, the second coordinates give $t = 2$ and the first give $s + t = 0$, so $s = -2$.
3. The coordinate columns are the columns:

$$\mathcal{M}_B^C(T) = \begin{bmatrix} 2 & -2 \\ 0 & 2 \end{bmatrix}.$$

**Worked example: null space and range of a matrix.** Let $A$ be the matrix below, viewed as a map $\mathbb{R}^3 \to \mathbb{R}^2$.

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 0 \end{bmatrix}$$

1. $Ax = 0$ is $x_1 + 2x_2 + x_3 = 0$ and $2x_1 + 4x_2 = 0$. The second equation gives $x_1 = -2x_2$; then the first gives $x_3 = 0$. So $\operatorname{null} A = \operatorname{span}\big((-2, 1, 0)\big)$, of dimension 1.
2. range $A$ is the span of the columns $(1, 2), (2, 4), (1, 0)$. The first and third are independent, so range $A = \mathbb{R}^2$, of dimension 2.
3. Check: $1 + 2 = 3 = \dim \mathbb{R}^3$.

**Examples 30 and 31.** **(30)** Multiplying each row by the column,

$$\begin{bmatrix}1 & 2 & 3\\ 2 & 3 & 4\end{bmatrix}\begin{bmatrix}5\\7\\9\end{bmatrix} = \begin{bmatrix}5+14+27\\10+21+36\end{bmatrix} = \begin{bmatrix}46\\67\end{bmatrix}.$$

**(31)** For the $T$ of Ex. 28, $T(1+2x+x^2)$: multiply $\mathcal{M}(T)$ by $(1,2,1,0)^{\mathsf{T}}$ to get $(4, 2i, 0, -2i)$. Check: $p(1) = 4$, $p(i) = 1 + 2i - 1 = 2i$.

**Definition** (Def. 20 / LADR 3.43: matrix product)**.** For $T \in \mathcal{L}(U,V)$, $S \in \mathcal{L}(V,W)$ and bases $u_1,\dots,u_l$ of $U$, $v_1,\dots,v_m$ of $V$, $w_1,\dots,w_n$ of $W$: the product is defined so that $\mathcal{M}(S)\mathcal{M}(T) = \mathcal{M}(ST)$. In entries, if $A = (a_{ij})$ is $n \times m$ and $B = (b_{jk})$ is $m \times l$, then $(AB)_{ik} = \sum_{j=1}^{m} a_{ij}b_{jk}$.

**The idea**

Column $k$ of $\mathcal{M}(ST)$ is the coordinates of $S(T(u_k))$. The coordinates of $T(u_k)$ are column $k$ of $B$, and applying $S$ to a vector multiplies its coordinates by $A$. So *column $k$ of $AB$ is $A$ times column $k$ of $B$*. This is why the number of columns of $A$ must equal the number of rows of $B$: both are $\dim V$.

*Derivation of the entries of the product in Def. 20.* 1. Column $k$ of $\mathcal{M}(T)$ says $T(u_k) = \sum_j b_{jk}v_j$, and column $j$ of $\mathcal{M}(S)$ says $S(v_j) = \sum_i a_{ij}w_i$.

2. Apply $S$ and use its linearity:

$$ST(u_k) = S\Big(\sum_j b_{jk}v_j\Big) = \sum_j b_{jk}S(v_j) = \sum_j b_{jk}\sum_i a_{ij}w_i = \sum_i\Big(\sum_j a_{ij}b_{jk}\Big)w_i.$$

3. The coefficient of $w_i$ is the $(i,k)$ entry of $\mathcal{M}(ST)$, so $(AB)_{ik} = \sum_j a_{ij}b_{jk}$. This formula uses only the entries of $A$ and $B$, so $AB$ does not depend on which maps and bases produced $A$ and $B$.

□

In the same way, row $i$ of $AB$ is (row $i$ of $A$) $B$, since $(AB)_{ik} = \sum_j a_{ij}b_{jk}$ uses only row $i$ of $A$.

**Example 32: antiderivative after derivative.** $T$ = differentiation $\mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$, $S$ = antiderivative with zero constant term $\mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$, standard bases. Multiplying the two matrices,

$$\mathcal{M}(S)\mathcal{M}(T) = \begin{bmatrix}0 & 0 & 0\\ 1 & 0 & 0\\ 0 & 1/2 & 0\\ 0 & 0 & 1/3\end{bmatrix}\begin{bmatrix}0 & 1 & 0 & 0\\ 0 & 0 & 2 & 0\\ 0 & 0 & 0 & 3\end{bmatrix} = \begin{bmatrix}0 & 0 & 0 & 0\\ 0 & 1 & 0 & 0\\ 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 1\end{bmatrix},$$

so $ST$ removes the constant term.

**Matrix times column is a special case of matrix times matrix.** For $v \in V$, let $T_v\colon \mathbb{F} \to V$ be the linear map with $T_v(1) = v$. In the basis 1 of $\mathbb{F}$, the single column of $\mathcal{M}(T_v)$ is $\mathcal{M}(v)$. So

$$\mathcal{M}(S)\,\mathcal{M}(v) = \mathcal{M}(S)\,\mathcal{M}(T_v) = \mathcal{M}(ST_v) = \mathcal{M}(S(v)),$$

which is Def. 19 again.

**Proposition** (Prop. 27: rules for matrix products)

For matrices of compatible sizes: $(AB)C = A(BC)$, $A(B+C) = AB + AC$, $(A+B)C = AC + BC$, and $AI = A$, $IB = B$, where $I = \mathcal{M}(\mathrm{id}_V)$ is the square matrix with 1 on the diagonal and 0 elsewhere (column $j$ holds the coordinates of $\mathrm{id}(v_j) = v_j$).

*Proof.* 1. Give each $\mathbb{F}^k$ its standard basis. By Prop. 26 (surjectivity), every matrix is $\mathcal{M}$ of some linear map. So write $A = \mathcal{M}(S)$, $B = \mathcal{M}(T)$, $C = \mathcal{M}(R)$, and in the distributive laws $C = \mathcal{M}(T')$, resp. $B = \mathcal{M}(S')$.

2. The product depends only on the entries (derivation after Def. 20), so $AB = \mathcal{M}(ST)$ for these particular maps.
3. **Associativity.** $(AB)C = \mathcal{M}(ST)\mathcal{M}(R) = \mathcal{M}((ST)R)$ and $A(BC) = \mathcal{M}(S)\mathcal{M}(TR) = \mathcal{M}(S(TR))$. These are equal because $(ST)R = S(TR)$.
4. **Left distributive law.** Using Prop. 26 for the sums,

$$A(B+C) = \mathcal{M}(S)\mathcal{M}(T+T') = \mathcal{M}(ST+ST') = \mathcal{M}(ST) + \mathcal{M}(ST') = AB + AC.$$

5. **Right distributive law.** In the same way,

$$(A+B)C = \mathcal{M}(S+S')\mathcal{M}(R) = \mathcal{M}(SR+S'R) = AC + BC.$$

6. **Identity.** $AI = \mathcal{M}(S)\mathcal{M}(\mathrm{id}) = \mathcal{M}(S\,\mathrm{id}) = A$, and $IB = \mathcal{M}(\mathrm{id}\,T) = B$.

□

**Permutation matrices.** For a permutation $\sigma$ of $\{1,\dots,n\}$, let $S_\sigma e_j = e_{\sigma(j)}$. Column $j$ of its matrix is $e_{\sigma(j)}$, so each column has exactly one 1. Row $i$ has its 1 only in column $j = \sigma^{-1}(i)$, because $\sigma$ is a bijection; so each row has exactly one 1. Since $S_\sigma S_\tau e_j = S_\sigma e_{\tau(j)} = e_{\sigma(\tau(j))}$, the maps $S_\sigma S_\tau$ and $S_{\sigma\tau}$ agree on a basis, so $S_\sigma S_\tau = S_{\sigma\tau}$ (Prop. 21). Induction on $k$ gives $S_\sigma^k = S_{\sigma^k}$. Instance: for $\sigma(1)=2$, $\sigma(2)=3$, $\sigma(3)=1$, $S_\sigma$ is the matrix below, and $S_\sigma^3 = S_{\sigma^3} = I$.

$$S_\sigma = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

**Vectors as representations** (notes' remark): in machine learning a token is a vector $x \in \mathbb{R}^d$ and a learned matrix $W$ acts by $x \mapsto Wx$. This is an ordinary linear map; the dimension is just large.

**The question answered.** Column $j$ of $\mathcal{M}(T)$ holds the coordinates of the image of the $j$-th basis vector, and the table of a composition is the product of the tables: $\mathcal{M}(SD) = \mathcal{M}(S)\mathcal{M}(D)$, computed in Ex. 32.

### Shortcuts

- **Matrix of a map.** Column $j$ holds the coordinates of $T(v_j)$ in the target basis. Reason: Def. 17. Condition: when the target basis is not standard, solve for the coordinates. Instance: the worked example in non-standard bases.
- **Range: the span of the columns. Null space: solve $Ax = 0$.** Reason: the idea box after Def. 19. Instance: the worked example on null space and range.
- **One column or one row of a product.** Column $k$ of $AB$ is $A$ times column $k$ of $B$; row $i$ of $AB$ is row $i$ of $A$ times $B$. Reason: $(AB)_{ik} = \sum_j a_{ij}b_{jk}$. Instance: column 2 of the product in Ex. 32 is $\mathcal{M}(S)$ times $(1,0,0)^\mathsf{T}$, which is the first column of $\mathcal{M}(S)$, namely $(0,1,0,0)^\mathsf{T}$.
- **Composition is multiplication in the same order.** $\mathcal{M}(ST) = \mathcal{M}(S)\mathcal{M}(T)$: the map applied first is on the right. Reason: Def. 20. Instance: Ex. 32.

### Traps

- The matrix depends on the bases, including their order. For $N$ in the basis $e_2, e_1$: $N(e_2) = e_1$ is the second basis vector and $N(e_1) = 0$, so the matrix is $\left[\begin{smallmatrix} 0 & 0 \\ 1 & 0 \end{smallmatrix}\right]$, not $\left[\begin{smallmatrix} 0 & 1 \\ 0 & 0 \end{smallmatrix}\right]$.
- Coordinates are not the vector: in Ex. 29, $(0,1,2)$ is the coordinate column of $1+2x+3x^2$ in the basis $1+x, 1+x^2, x+x^2$, not the polynomial $x+2x^2$.
- $AB$ can be defined while $BA$ is not: for $A \in \mathbb{F}^{2,3}$ and $B \in \mathbb{F}^{3,1}$, $AB \in \mathbb{F}^{2,1}$, but $BA$ would need $1 = 3$.
- Apply the given formula to the given basis vectors; do not choose the images. In Practice Midterm 1 Q3, $T(1,0,0) = 1+x^2$, not 1.

## §2.4 Stochastic matrices and Markov chains (Ch. 6.3)

**The question.** A system is in one of the states A, B, C. From A it stays in A or moves to B, each with probability $\frac{1}{2}$. From B it moves to A, B, C with probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. From C it moves to B or stays in C, each with probability $\frac{1}{2}$. It starts in A. What are the probabilities of the three states after three steps?

- A **probability vector** $p \in \mathbb{R}^n$ has $p_j \geq 0$ and $\sum p_j = 1$; $p_j$ is the probability of being in state $j$. Example: with states A, B, C, $p = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$ says the system is in B with probability $\frac{1}{2}$.
- $P \in \mathbb{R}^{n,n}$ is **stochastic** if every entry is $\geq 0$ and every *column* sums to 1. Column $j$ is the distribution of the next state, given the present state is $j$ (it is $Pe_j$).
- If the present distribution is $p$, the next one is $Pp$: $(Pp)_i = \sum_j p_{ij} p_j$ adds, over present states $j$, (probability of being in $j$) $\times$ (probability of moving from $j$ to $i$). $Pp$ is again a probability vector: its entries are $\geq 0$, and

$$\sum_i (Pp)_i = \sum_j \Big( \sum_i p_{ij} \Big) p_j = \sum_j p_j = 1.$$

- A **Markov chain** applies the same $P$ repeatedly: $p_k = P^k p_0$. The **Markov assumption** is that the next step depends only on the present state, not on the earlier history. Other books use row vectors and row-stochastic matrices; that is the transpose of this convention.
- A permutation matrix is stochastic: from state $j$ the next state is certainly $\sigma(j)$.

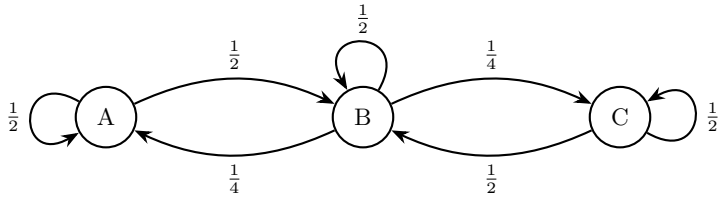
**Example 33: a three-state Markov chain.** The chain of the question has the stochastic matrix below: column A says that from A the chain stays in A or moves to B with probability $\frac{1}{2}$ each; column B says that from B it moves to A, B, C with $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$; column C likewise.

$$P = \begin{bmatrix} 1/2 & 1/4 & 0 \\ 1/2 & 1/2 & 1/2 \\ 0 & 1/4 & 1/2 \end{bmatrix}$$

Start from $p_0 = (1, 0, 0)$. The next three distributions are

$$\begin{aligned} p_1 &= Pp_0 = (\tfrac{1}{2}, \tfrac{1}{2}, 0) \qquad \text{(the first column of } P\text{)}, \\ p_2 &= Pp_1 = (\tfrac{3}{8}, \tfrac{1}{2}, \tfrac{1}{8}), \\ p_3 &= Pp_2 = (\tfrac{5}{16}, \tfrac{1}{2}, \tfrac{3}{16}). \end{aligned}$$

For example, $(p_2)_1 = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{4} \cdot \frac{1}{2} + 0 \cdot 0 = \frac{3}{8}$. The distributions approach $\pi = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$, and $P\pi = \pi$: once the chain has distribution $\pi$, its distribution does not change. Why $P^k p_0$ approaches $\pi$, and how fast, is explained in a later week.




The chain of Ex. 33. The arrows out of each state carry column $j$ of $P$; they add up to 1.

**The question answered.** $p_3 = P^3 p_0 = (\frac{5}{16}, \frac{1}{2}, \frac{3}{16})$.

**Shortcuts**

- **Is $P$ stochastic?** Check that every entry is $\geq 0$ and every column sums to 1. Reason: the definition. Instance: in Ex. 33 the columns sum to $\frac{1}{2} + \frac{1}{2}$, $\frac{1}{4} + \frac{1}{2} + \frac{1}{4}$, $\frac{1}{2} + \frac{1}{2}$.

- **Is $\pi$ stationary?** Check $P\pi = \pi$ and that $\pi$ is a probability vector. Instance: in Ex. 33,

$$P\begin{bmatrix}1/4\\1/2\\1/4\end{bmatrix} = \begin{bmatrix}1/8+1/8\\1/8+1/4+1/8\\1/8+1/8\end{bmatrix} = \begin{bmatrix}1/4\\1/2\\1/4\end{bmatrix}.$$

- **First step from a pure state.** $Pe_j$ is column $j$. Reason: $Ae_j$ is column $j$ of $A$. Instance: $p_1 = Pe_1 = (\frac{1}{2}, \frac{1}{2}, 0)$ in Ex. 33.

**Traps**

- Stochastic here means columns sum to 1, not rows. $Q = \left[\begin{smallmatrix}1&0\\1&0\end{smallmatrix}\right]$ has rows summing to 1, but $Q(1,0)^{\mathsf{T}} = (1,1)^{\mathsf{T}}$ is not a probability vector.
- Books that use row vectors use the transpose $P^{\mathsf{T}}$ and multiply on the left. Do not mix the conventions within one computation: with the matrix of Ex. 33, $P^{\mathsf{T}}(1,0,0)^{\mathsf{T}} = (\frac{1}{2}, \frac{1}{4}, 0)^{\mathsf{T}}$, which sums to $\frac{3}{4}$.

## §3 Problem set

**HW3 P1.** Let $T\colon V \to W$ be linear, $V$ finite-dimensional, $U$ a subspace of $W$.

(a) Prove that $T^{-1}(U) = \{v \in V : Tv \in U\}$ is a subspace of $V$.

(b) Prove $\dim T^{-1}(U) = \dim \operatorname{null} T + \dim(U \cap \operatorname{range} T)$.

(c) Let $D$ be differentiation on $\mathcal{P}_4(\mathbb{R})$. Using (b), find the dimension of the space of $p$ with $p'(x) = ax^2 + b$ for some $a, b \in \mathbb{R}$.

*Hint:* apply rank–nullity to the restriction of $T$ to $T^{-1}(U)$.

**HW3 P2.** (a) Find a linear map $T\colon \mathbb{R}^4 \to \mathbb{R}^4$ with $\operatorname{null} T = \operatorname{range} T$. (b) Prove that no linear map $T\colon \mathbb{R}^5 \to \mathbb{R}^5$ satisfies $\operatorname{null} T = \operatorname{range} T$.

*Hint:* rank–nullity; model (a) on $N$.

**HW3 P3.** Let $U = \{p \in \mathcal{P}_4(\mathbb{R}) : p(1) = p'(1) = 0\}$. Define a linear map $S\colon \mathcal{P}_4(\mathbb{R}) \to \mathbb{R}^2$ with null space $U$ and use rank–nullity to find $\dim U$. Compare with Homework 2.

*Hint:* $S(p) = (p(1), p'(1))$; show it is surjective.

**HW3 P4.** Let $T\colon V \to W$ be linear. (a) Prove that if $v_1, \dots, v_n$ is dependent, so is $Tv_1, \dots, Tv_n$. (b) Give an example where an independent list is not sent to an independent list.

*Hint:* apply $T$ to a relation.

**HW3 P5.** Define $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_4(\mathbb{R})$ by $(Tp)(x) = x^2 p'(x) + p(x)$. (a) Find the matrix of $T$ in the bases $1, x, x^2, x^3$ and $1, x, x^2, x^3, x^4$. (b) Determine $\operatorname{null} T$ and $\operatorname{range} T$ and their dimensions. (c) Explain from (a) why the range is the span of the columns.

*Hint:* column $j$ is the image of the $j$-th basis vector.

**HW3 P6.** Solve $x + y + z + w = 1$, $x + z = 0$ over $\mathbb{F}_2$. List all solutions.

*Hint:* $-1 = 1$ and $2 = 0$.

**HW3 P7 (optional).** Let $A$ be an $n \times m$ matrix over $\mathbb{F}_2$ with $Ax = b$ consistent. Express the number of solutions in terms of $m$ and $\operatorname{rank} A$.

*Hint:* if $x_0$ is one solution of $Ax = b$, the solutions are exactly $x_0 + y$ with $y \in \operatorname{null} A$, and a $k$-dimensional space over $\mathbb{F}_2$ has $2^k$ elements.

**R1 (Practice Midterm 1, Question 3).** Define $T\colon \mathbb{R}^3 \to \mathcal{P}_2(\mathbb{R})$ by $T(a,b,c) = (a+b) + (b+c)x + (a+c)x^2$. Use the basis $B = (1,0,0), (1,1,0), (1,1,1)$ of the domain and $C = (1, x, x^2)$ of the codomain. (a) Find $\mathcal{M}(T)$. (b) Prove that $T$ is bijective.

*Hint:* column $j$ holds the coordinates, in $1, x, x^2$, of $T$ applied to the $j$-th vector of $B$; for (b), $\dim \mathbb{R}^3 = \dim \mathcal{P}_2(\mathbb{R})$, so injectivity is enough (Prop. 25).

**R2 (Practice Midterm 1, Question 4d).** Let $D\colon \mathcal{P}_4(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation and $U = \operatorname{span}(1, x^2) \subseteq \mathcal{P}_3(\mathbb{R})$. Use HW3 P1(b) to find $\dim D^{-1}(U)$, and give a basis of $D^{-1}(U)$.

*Hint:* $\operatorname{null} D = \operatorname{span}(1)$ and $\operatorname{range} D = \mathcal{P}_3(\mathbb{R})$; for the basis, take a basis of $\operatorname{null} D$ and one preimage under $D$ of each basis vector of $U$.

**R3 (Midterm 1 review, Exercise 2).** Let $T\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_2(\mathbb{R})$, $T(p) = p'' + p'$. Compute $\mathcal{M}(T)$ in the standard bases.

*Hint:* apply $T$ to $1, x, x^2, x^3$ and write each image in the basis $1, x, x^2$.

**R4 (Midterm 1 review, Exercise 6).** Does there exist a linear map $T\colon \mathbb{R}^4 \to \mathbb{R}^2$ with $\operatorname{null} T = \{(x,x,x,x) : x \in \mathbb{R}\}$?

*Hint:* rank–nullity, and $\dim \operatorname{range} T \le \dim \mathbb{R}^2$.

**R5 (Midterm 1 review, Exercise 4).** Let $S\colon \mathcal{P}_2(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$, $S(p) = \int_0^x p(t)\,dt - p'(x)$. (i) Argue that $S$ cannot be surjective. (ii) Find $\operatorname{null} S$ and $\operatorname{range} S$. (iii) Find the matrix of $S$ in the standard bases.

*Hint:* (i) $\dim \operatorname{range} S \le \dim \mathcal{P}_2(\mathbb{R})$; (iii) apply $S$ to $1, x, x^2$.

**R6 (Midterm 1 review, Exercise 11).** Give an example of $T \in \mathcal{L}(\mathbb{R}^2)$ with $\mathbb{R}^2 = \ker T \oplus \operatorname{range} T$.

*Hint:* Ex. 23 and the last bullet after Prop. 25.